

Solutions

If $R_{\alpha\beta\gamma\delta} = K(g_{\alpha\gamma}g_{\beta\delta} - g_{\alpha\delta}g_{\beta\gamma})$ represents a constant curvature space-time region, then

1. show that K is constant. [3]

$$R_{\alpha\beta\gamma\delta} = K(g_{\alpha\gamma}g_{\beta\delta} - g_{\alpha\delta}g_{\beta\gamma})$$

0.5 pt for finding $R_{\beta\gamma\delta}^{\mu}$:

$$g^{\mu\alpha}R_{\alpha\beta\gamma\delta} = R_{\beta\gamma\delta}^{\mu} = K(g^{\mu\alpha}g_{\alpha\gamma}g_{\beta\delta} - g^{\mu\alpha}g_{\alpha\delta}g_{\beta\gamma}) = K(\delta_{\gamma}^{\mu}g_{\beta\delta} - \delta_{\delta}^{\mu}g_{\beta\gamma})$$

1 pt for finding $R_{\beta\delta}$:

$$R_{\beta\delta} = R_{\beta\mu\delta}^{\mu} = K(4g_{\beta\delta} - g_{\beta\delta}) = 3Kg_{\beta\delta}$$

1 pt for finding Ricci scalar R :

$$g^{\beta\delta}R_{\beta\delta} = 12K$$

0.5 pt for saying that K is constant since R is constant.

2. show that if such a space-time satisfies Einstein's equations then

$$K = -\frac{2\pi G}{3c^4}T$$

.

[2]

0.5 pt for writing Einstein's equations

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = \frac{8\pi G}{c^4}T_{\mu\nu}$$

0.5 pt for contracting with $g^{\mu\nu}$

$$R - \frac{1}{2}(4)R = \frac{8\pi G}{c^4}T$$

1 pt for substituting R from above and simplifying to get correct answer.

$$\begin{aligned} -12K &= \frac{8\pi G}{c^4}T \\ K &= -\frac{2\pi G}{3c^4}T \end{aligned}$$